

Design a Unique ID Generator - UUID Approach

1. What is UUID?

- UUID = **Universally Unique Identifier**
- A **128-bit number** used to uniquely identify data

Key Property

- Extremely low probability of collision

Quote:

None

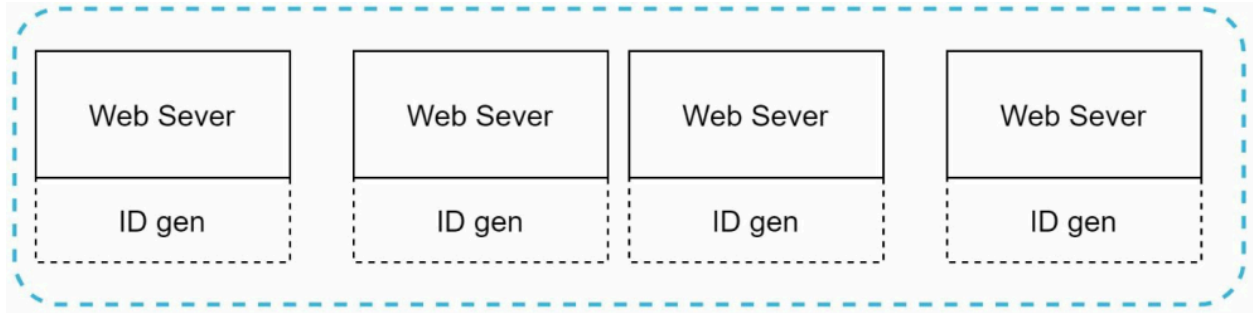
Generating 1 billion UUIDs per second for ~100 years → 50% chance of one duplicate

Example

None

09c93e62-50b4-468d-bf8a-c07e1040bfb2

2. How It Works



- Each **web server generates IDs independently**
- No coordination between servers required

3. Pros

- Simple to generate
- No synchronization required
- Highly scalable (scales with number of servers)

4. Cons

- IDs are **128-bit**, but requirement is **64-bit**
- Not **time-ordered**
- May be **non-numeric**

5. Key Takeaway

UUID:

- Easy and scalable
- But does not satisfy constraints like **64-bit, ordering, numeric-only**